

HERB: 一个高通量实验与参考指导的中药数据库

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摘要

药物转录组学已成为评估药物治疗效果和发现新药物靶点的有力方法。近年来, 传统中药 (TCM) 的研究越来越多地采用高通量转录组筛选来研究草药成分的分子作用。大量研究已经探讨了草药成分的基因靶点, 并将草药成分与各种现代疾病联系起来。然而, 目前尚无系统的数据库来组织这些中药数据。因此, 我们建立了 HERB, 一个以高通量实验和参考资料为指导的中药数据库, 其中文名称为“本草纵鉴”。我们重新分析了来自1037个高通量实验的6164个基因表达谱, 这些实验评估了中药草药成分, 并通过将HERB中的综合药物转录组学数据映射到CMap (最大的现代药物数据集), 生成了中药草药成分与2837种现代药物之间的关联。

中医药的现代化, 并指导合理的现代药物研发工作。可通过 <http://herb.ac.cn/> 访问。

介绍

对化学化合物分子效应的全面理解有助于在现代药物发现中战略性地选择候选化合物 (1)。为了评估活性化合物的分子效应, 常常使用细胞系和动物模型的功能性实验, 通过高通量技术研究全转录组的变化, 以识别各种处理或扰动的影响 (2)。现代药物中最大的此类数据库称为 Connectivity Map (CMap), 其中包含数千种经过良好注释的小分子在核心九种细胞系中进行转录组水平扰动的数据集 (3)。此外, 还有其他类似的数据可从存储功能基因组学数据集的数据库获得, 例如基因表达综合数据库 (GEO) (4)。最近此类转录组扰动数据集的广泛可用性彻底改变了药理学领域, 并帮助研究人员快速鉴定

HERB: a high-throughput experiment- and reference-guided database of traditional Chinese medicine

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ABSTRACT

Pharmacotranscriptomics has become a powerful approach for evaluating the therapeutic efficacy of drugs and discovering new drug targets. Recently, studies of traditional Chinese medicine (TCM) have increasingly turned to high-throughput transcriptomic screens for molecular effects of herbs/ingredients. And numerous studies have examined gene targets for herbs/ingredients, and link herbs/ingredients to various modern diseases. However, there is currently no systematic database organizing these data for TCM. Therefore, we built HERB, a high-throughput experiment- and reference-guided database of TCM, with its Chinese name as Ben-CaoZuJian. We re-analyzed 6164 gene expression profiles from 1037 high-throughput experiments evaluating TCM herbs/ingredients, and generated connections between TCM herbs/ingredients and 2837 modern drugs by mapping the comprehensive pharmacotranscriptomics dataset in HERB to CMap, the largest such dataset for modern drugs. Moreover, we manually curated 1241 gene targets and 494 modern diseases for 473 herbs/ingredients from 1966 references published recently, and cross-referenced this novel information to databases containing such data for drugs. Together with database mining and statistical inference, we linked 12 933 targets and 28 212 diseases to 7263 herbs and 49 258 ingredients and provided six pairwise relationships among them in HERB. In summary, HERB will intensively support

the modernization of TCM and guide rational modern drug discovery efforts. And it is accessible through <http://herb.ac.cn/>.

INTRODUCTION

Having a complete understanding of the molecular effects of chemical compounds facilitates strategic selection of candidates for advancement in modern drug discovery (1). To evaluate the molecular effects of active compounds, functional assays using cell lines and animal models are often utilized to study whole-transcriptomic changes by using high-throughput technologies to identify the effects of various treatments or perturbations (2). The largest such database for modern drugs, called Connectivity Map (CMap), includes transcriptomic-level perturbation datasets for thousands of well-annotated small molecules profiled in a core set of nine cell lines (3). Furthermore, there are other similar data available from databases storing functional genomics datasets, e.g. the Gene Expression Omnibus (GEO) repository (4). The recent explosion of availability of such transcriptomics perturbation datasets has transformed the field of pharmacology and helped researchers rapidly identify promising chemical compounds for various diseases (5). For example, researchers recently identified a compound, celastrol, which acts as a leptin sensitizer to treat obesity, by mapping the gene expression profile of celastrol to that of reduced endoplasmic reticulum (ER) stress, a condition tightly linked to obesity (6). Similarly, withaferin A, another leptin sensitizer for the treatment of obesity, was discovered through a CMap library analysis of those small molecules that have gene expression profiles similar to that of celastrol (7).

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传统中医 (TCM) 为现代药物发现和开发提供了丰富的基础。截至目前, 根据DrugBank的记录 (补充表 S1), 已有97种经FDA批准的药物源自中医。例如, 麻黄 (Herba ephedrae) 提取的麻黄碱被用作抗哮喘药。同样, 从中药青蒿 (Herba Artemisiae annuae) 中提取的青蒿素 (qinghaosu), 现已成为治疗疟疾的一线药物。中草药的传统知识及其组方是建立在几千年的民间试验基础上的, 包括通过迭代的方法识别对多种疾病具有临床疗效的天然产物。因此, 分离中医药中的活性成分并进一步解析其作用机制, 为新型治疗药物提供了有前景的起点。

随着下一代测序技术的出现, 越来越多的中医研究开始利用高通量技术识别活性草药及成分的分子作用。例如, 自2006年和2011年以来, 微阵列和RNA测序方法分别被用于中医研究。截至2019年, 在GEO数据库中, 有 >6000 个样本涉及经中药/成分处理的细胞系、动物或患者, 然后使用这些转录组技术进行分析。与天然产物相关的组学数据的快速积累, 为理解每种草药/成分活性引起的转录响应和调控变化提供了新的机会。重要的是, 这些数据可以映射到CMap, 以系统评估中药成分与现代药物之间的相似性, 为中医治疗作为候选疗法的研究提供客观、数据驱动的支持。

最近已经发表了大量研究, 旨在解码中药活性成分的基因靶点, 并将其与现代疾病联系起来。最近对这些参考文献进行整理的工作是分别于2011年和2012年发布的HIT (15) 和TCMID (16) 数据库。HIT从PubMed挖掘中收集了586种草药成分相关的1,301个基因靶点, 而TCMID则包含从WeiPu数据库的中文文献中挖掘的680个草药靶点。然而, 以往从未系统收集和整理过草药-靶点、草药-疾病及成分-疾病之间的典型关联。经过整理的信息其价值明显高于以往通过中间成分 (如靶点) 间接关联的努力。因此, 过去十年发表的大部分研究都需要人工整理, 以开发一个高可信度的数据库, 将靶点、疾病与中药草药及成分关联起来。

因此, 我们构建了 HERB (本草组建作为其中文名), 一个高通量实验和参考

中医指南数据库。HERB 整合了多个中医数据库, 因此包含迄今为止最全面的草药及成分清单。我们收集并统一重新分析了所有可用高通量实验 (即微阵列和RNA-seq 实验) 中测试草药或其活性成分的公开数据。通过这种方式, 我们生成了一个包含来自 1037 个实验的6164 个基因表达谱的库, 用于评估草药或成分, 并基于数据构建了草药、活性成分与 CMap 药物转录组数据集中 2837 种现代药物之间的关联。此外, 我们通过PubMed 文本挖掘收集了自 2011 年以来发表的 17,886 篇与中医相关的论文, 并手动关联了 1241 个基因靶点和 494 种现代疾病, 对应于 473 种草药/成分, 来源于其中 1966 条参考文献。中医这一新整理的靶标和疾病信息由于经过手动确认, 因此具有高质量, 并与数据库进行了交叉比对。

材料与方法

中医药材和成分目录

在HERB中, 我们首先准备了一份草药及成分清单, 并通过整合多个中医药数据库 (包括SymMap (21)、TCMID 2.0 (22)、TCMSP 2.3 (23) 和 TCM-ID (24)) 来确定它们之间的关系。为了获得不重复的清单, 我们将具有相同ID、名称或别名的草药和成分合并。对于数据库之间的不一致情况, 我们选择最新发布的条目。需要注意的是, 所有成分的分子式和分子SMILES信息均依据权威化学数据库SciFinder (25) 和PubChem (26) 进行了标准化。随后, 我们将中医药成分映射到DrugBank数据库 (27), 并将其中一部分中医药成分标注为已批准药物。最后, 我们检索了一个权威的活性草药成分数据库——“中药化学成分全国数据库” (<http://cintmed.cintcm.com/cintmed/>), 该数据库由中医药信息研究所持续维护了数十年。

中药草药及成分的高通量实验数据

我们收集了来自 GEO 的公开数据, 这些数据来自研究多种细胞类型或动物模型中草药或成分处理的高通量实验。我们只保留了那些在人类 (Homo sapiens) 或小鼠 (Mus musculus) 中生成, 并且通过典型的批量 RNA 测序平台 (Illumina) 或微阵列研究获得的数据集。随后, 我们建立了不同的流程, 用于统一重新分析 RNA 测序和微阵列数据。

Traditional Chinese medicine (TCM) provides a rich basis for modern drug discovery and development. To date, 97 FDA-approved drugs have been derived from TCM, as recorded by DrugBank (Supplementary Table S1). For example, ephedrine, which is derived from the TCM herb Ma Huang (*Herba ephedrae*), is used as an anti-asthmatic drug (8). Similarly, artemisinin (qinghaosu), derived from the Chinese herb qinghao (*Herba Artemisiae annuae*), is now a first-line drug for malaria (9). The traditional knowledge of Chinese herbs, as well as their combinations as prescriptions, is built upon thousands of years of folk testing, including iteratively identifying natural products with improved clinical efficacy for the treatment of a wide range of diseases. Therefore, isolating the active ingredients in TCM and further dissecting their mechanisms of action provides a promising starting point for new therapeutics (10,11).

With the advent of next-generation sequencing technology, an increasing number of TCM studies have focused on identifying the molecular effects of active herbs and ingredients using high-throughput techniques (12). For example, microarray and RNA sequencing approaches have been used in TCM research since 2006 and 2011, respectively. Through 2019, there were >6000 samples in the GEO database on cell lines, animals, or patients that were treated with TCM herbs/ingredients and then analyzed using these transcriptomic technologies. The rapid accumulation of omics data related to natural products has provided new opportunities to understand the transcriptional responses and regulatory changes resulting from the activity of each herb/ingredient (13). Importantly, these data can be mapped to CMap to systematically evaluate the similarities between TCM components and modern drugs, providing objective, data-driven support for the investigation of TCM therapies as candidates for novel therapeutics, such as TCM-HUB (14). However, due to the diversity of data resulting both from varying production platforms and experimental conditions, there is currently no systematic database organizing the data and findings from these high-throughput TCM experiments.

There have been a large number of studies recently published aiming to decode gene targets of the active components of TCM, and link them to modern diseases. The most recent efforts to curate these references were the HIT (15) and TCMID (16) databases published in 2011 and 2012, respectively. HIT contains 1,301 gene targets related to 586 herbal ingredients gathered from PubMed mining, and TCMID contains 680 herbal targets from Chinese articles mined from the WeiPu database. However, the typical sets of associations between herb-target, herb-disease, and ingredient-disease, have not previously been collected and curated from published references. Curated information is of significantly higher value than previous efforts that indirectly linked them using intermediate components, such as targets. Thus, the bulk of studies published within the last decade required manual curation in order to develop a high-confidence database linking targets, diseases, and TCM herbs/ingredients. Such a system would offer high-quality, evidence-based connections between TCM and modern drugs.

Therefore, we built **HERB** (BenCaoZuJian as its Chinese name), a high-throughput experiment- and reference-

guided database of TCM. HERB integrates multiple TCM databases and thus contains the most comprehensive list of herbs and ingredients created to date. We gathered and uniformly reanalyzed public data from all available high-throughput experiments (i.e. microarrays and RNA-seq experiments) that tested herbs or their active ingredients. In this manner, we generated a library containing 6164 gene expression profiles from 1037 experiments evaluating herbs or ingredients and built data-driven connections among herbs, active ingredients, and 2837 modern drugs with pharmacotranscriptomics datasets in CMap. Furthermore, we collected 17,886 TCM-related papers that were published since 2011 by PubMed text mining and manually connected 1241 gene targets and 494 modern diseases for 473 herbs/ingredients from 1966 of those references. This newly curated target and disease information for TCM is of high quality due to its manual confirmation, and was cross-referenced to databases for modern drugs, including the TTD (17), DisGeNet (18), HPO (19), and Disease Ontology (20) databases. Together with database mining and statistical inference, we linked 12 933 targets and 28 212 diseases to 7263 herbs and 49 258 ingredients within HERB. The objective, data-based connections between TCM and modern medicines described in HERB provide strong support for further pharmacological studies of TCM as a fundamental arm of modern drug discovery efforts.

MATERIALS AND METHODS

Catalogs of TCM herbs and ingredients

In HERB, we first prepared a list of herbs and ingredients and determined their relationships by integrating multiple TCM databases including SymMap (21), TCMID 2.0 (22), TCMSP 2.3 (23) and TCM-ID (24). To obtain a non-redundant list, we merged herbs/ingredients with the same IDs, names, or aliases. In case of discrepancies across databases, we selected the entry with the most recent publication date. Note that the molecule formula and molecule smile information for all ingredients were standardized according to SciFinder (25) and PubChem (26), two authoritative chemical databases. We then mapped the TCM ingredients to the DrugBank database (27) and labeled a subset of TCM ingredients as approved drugs. Last, we searched an authoritative database of active herbal ingredients, the ‘National Database for Chemical Composition in TCM’ (<http://cintmed.cintcm.com/cintmed/>), which has been continuously maintained for several decades by the Institute of Information on Traditional Chinese Medicine, China Academy of Chinese Medical Sciences.

High-throughput experimental data for TCM herbs and ingredients

We gathered public data from GEO high-throughput experiments that studied treatments with herbs or ingredients in multiple cell types or animal models. We only retained those datasets that were generated in *Homo sapiens* or *Mus musculus* and that were obtained by a typical bulk RNA sequencing platform (Illumina) or microarray studies. Then, we built different pipelines for uniformly reanalyzing RNA-seq and microarray data.

对于RNA-seq数据, 我们首先使用fasterq-dump 2.9.2 (28) 下载每个样本的fastq格式原始读取数据 (每个样本在GEO中具有唯一的GSM编号)。然后, 我们使用Trim Galore 0.6.4 (http://www.bioinformatics.babraham.ac.uk/projects/trim_galore/) 过滤接头序列和低质量读取, 参数为 ‘-quality 20 -length 20 -consider_already_trimmed 10’。接下来, 我们使用STAR 2.7.3a (29) 将过滤后的读取比对到人类基因组 (组装版本hg38, 注释v100) 或小鼠基因组 (组装版本mm10, 注释v100)。映射率低于60%的样本被丢弃。最后, 我们使用featureCounts 2.0.0 (30) 对每个样本的每个基因进行读取计数, 并使用R包limma 3.42.2 (31) 中的 ‘voom’ 函数对数据进行预处理和分位数标准化, 生成基因表达矩阵。

对于微阵列数据, 我们重新分析了提供原始数据的样本。我们首先使用R包oligo 1.50.0 (32) 针对Affymatrix平台提取原始数据中的探针表达谱, 或使用R包limma 3.42.2 (31) 针对Agilent平台提取。接下来, 我们使用稳健多芯片平均算法 (rma) 对探针表达谱进行归一化。对于无法获取原始数据的其他样本, 我们直接使用其探针的归一化表达谱。最后, 我们通过系统构建来自不同芯片平台的探针与基因之间的转换关系, 将探针表达谱转化为归一化的基因表达矩阵。当多个探针可转换为同一个基因时, 选择表达水平的中位数探针来代表该基因。为此, 我们提供了一个名为probe2gene的自动转换脚本, 该脚本嵌入了从NCBI下载的105个平台的SOFT格式转换文件, 并能够自动完成转换。

差异表达分析与功能富集分析

我们对每种中药材/成分进行了差异表达分析。首先, 我们手动提取了样本之间的比较关系。在NCBI-GEO中, 一个具有唯一GSE编号的实验总是包含若干样本 (每个样本有唯一的GSM编号), 这些样本包括对照样本以及在不同生物/生理条件下接受每种中药材/成分处理的样本。我们将一个比较定义为在相同条件下进行的一组对照样本和处理样本, 这里称为中药实验 (EXP)。为了确保可重复性, 我们要求每个EXP中处理样本至少有两个生物学重复 (≥ 2 GSMs)。由于每种中药材/成分可能有一个或多个EXP, 我们对每个EXP单独进行后续分析, 然后再合并来自多个EXP的分析结果。需要注意的是, 在人类或小鼠进行的EXP数据会被合并并分别展示。

对于每个EXP, 我们使用R软件包limma 3.42.2进行了差异表达分析。简而言之, 我们首先使用 “lmFit” 函数从标准化的基因表达矩阵中拟合线性模型, 然后使用 “eBayes” 函数计算经验贝叶斯统计量。最后,

我们选择了表达差异足够的基因, 即 $|\log_2(\text{倍数变化})| \geq 0.5$ 且 $P \leq 0.05$, 作为差异表达基因 (DEGs)。基于每个EXP的DEG列表, 我们使用R包clusterProfiler 3.14.3 (33) 对每个EXP进行了进一步的功能富集分析。我们使用 ‘enrichGO’ 函数进行GO富集分析, 使用 ‘enrichKEGG’ 函数进行KEGG富集分析。当 $P \leq 0.05$ 时选择富集的GO项和KEGG通路。值得注意的是, 我们在每个EXP中分别对所有基因、上调基因和下调基因进行了单独的GO/KEGG分析。

然后, 我们合并了每种具有多个实验 (EXP) 的草药/成分的分析结果。我们首先将实验的初始双尾P值转换为两个单尾P值, $P/2$ 和 $1 - P/2$, 以考虑上调和下调。然后, 我们使用Fisher方法 (34) 合并每个基因的两个统一概率, 一个用于上调 (P-up), 另一个用于下调 (P-down)。其检验统计量 (χ^2) 服从卡方分布, 如下所示。

$$\chi^2 = -2 \sum_{i=1}^k \ln(p_i)$$

我们使用R包metaseqR (35) 中的 ‘fisher.method’ 函数来计算统计量及相应的P值。对于只有一个显著P值的基因 (即 $P\text{-up} < 0 < P\text{-down}$ 或 $P\text{-down} < 0 < P\text{-up}$), 将其保留在差异表达基因 (DEG) 列表中, 并通过评估哪个P值显著来确定其表达失调的方向。同样, 我们还要求基于那些在单个测试中存在显著差异的EXPs, 平均 $|\log_2(\text{fold change})| \geq 0 > 0.5$ 。基于每种药材/成分的差异表达基因的合并集合, 我们分别对所有基因、上调基因和下调基因进行了GO和KEGG富集分析。此外, 我们使用BH方法 (36) 分别调整了上调和下调P值, 以控制假阳性发现率 (FDR)。

基于数据的中药成分与现代药物的映射

一旦我们确定了中草药/成分的基因表达谱, 我们进一步通过将HERB-EXP映射到CMap来评估中草药/成分与现代药物之间的相似性, CMap中包含数千种注释完善的小分子的药理转录组学数据集。因此, 我们首先将每个EXP导出的差异表达基因 (DEG) 列表映射到CMap, 然后合并来自特定草药或成分的所有EXP的映射结果, 从而建立了数据驱动的关联。需要注意的是, 在H. sapiens或M. musculus中进行的EXP数据是分别映射和显示的。

对于每个实验 (EXP), 差异表达基因列表以批量查询模式提交到CMap网站 (<https://clue.io/query>)。需要注意的是, DEG列表中最多保留前300个基因 (5)。值得注意的是, 小鼠实验 (EXP) 的DEG基因首先通过使用R包homologene结合NCBI的转换表 (<ftp://ftp.ncbi.nih.gov/pub/HomoloGene/build68/>) 转换为人类同源基因 (补充表)

For RNA-seq data, we first downloaded the raw reads in fastq format for each sample (with unique GSM numbers in GEO) using fasterq-dump 2.9.2 (28). Then, we filtered adaptor sequences and low-quality reads by using Trim Galore 0.6.4 (http://www.bioinformatics.babraham.ac.uk/projects/trim_galore/) with the parameters ‘-quality 20 -length 20 -consider_already_trimmed 10’. Next, we mapped the filtered reads to the genome of *Homo sapiens* (assembly hg38 and annotation v100) or *Mus musculus* (assembly mm10 and annotation v100) using STAR 2.7.3a (29). Samples with a mapping rate lower than 60% were discarded. Finally, we quantified the read counts for each gene in each sample using featureCounts 2.0.0 (30), preprocessed and quantile-normalized the data into a gene expression matrix using the ‘voom’ function in the R package limma 3.42.2 (31).

For microarray data, we reanalyzed the samples with raw data provided. We first extracted the probe expression profile from the raw data using R package oligo 1.50.0 (32) for the Affymatrix platform, or using R package limma 3.42.2 (31) for the Agilent platform. Next, we normalized the probe expression profile with the robust multichip average algorithm (rma). For other samples without accessible raw data, we directly used their normalized expression profile for probes. Finally, we transformed the probe profile into the normalized gene expression matrix by systematically constructing the conversion relationships between probes from diverse array platforms to genes. When multiple probes could be converted to the same gene, the probe with the median expression level was selected to represent that gene. To that end, we provide an automatic conversion script named probe2gene, which embeds conversion files in SOFT format for 105 platforms downloaded from NCBI and automatically conducts the probe-to-gene conversion according to the SOFT files.

Differential expression analysis and functional enrichment analysis

We conducted differential expression analysis for each TCM herb/ingredient. First, we manually extracted the comparative relationship between samples. In NCBI-GEO, an experiment with a unique GSE number always contains several samples (with unique GSM numbers), and the samples included control samples as well as samples treated with each herb/ingredient in varying biological/physiological conditions. We defined a comparison as a set of control samples and treatment samples that were performed in the same condition, called an HERB experiment (EXP) hereafter. To ensure reproducibility, we required at least two biological replicates of treatment samples (≥ 2 GSMs) in each EXP. As there can be one or many EXPs for each herb/ingredient, we performed downstream analysis individually for each EXP and then merged the analyzed results from multiple EXPs later. Note that data from EXPs performed in *Homo sapiens* or *Mus musculus* were merged and displayed separately.

For each EXP, we performed differential expression analysis using the R package limma 3.42.2. In brief, we first fit linear models from the normalized gene expression matrix using the ‘lmFit’ function and then computed the empirical Bayes statistics using the ‘eBayes’ function. Finally,

we selected genes with sufficient expression difference, i.e. $|\log_2(\text{fold change})| \geq 0.5$ and $P \leq 0.05$, as differentially expressed genes (DEGs). Based on the DEG list for each EXP, we conducted further functional enrichment analysis for each EXP using the R package clusterProfiler 3.14.3 (33). We used the ‘enrichGO’ function for GO enrichment and the ‘enrichKEGG’ function for KEGG enrichment. Enriched GO terms and KEGG pathways were selected when $P \leq 0.05$. Of note, we conducted separate GO/KEGG analyses in each EXP for all genes, up-regulated genes, and down-regulated genes, respectively.

Then, we merged the analyzed results for each herb/ingredient that had multiple EXPs. We firstly transformed the initial two-tailed P-values for EXPs to two one-tailed P -values, $P/2$ and $1 - P/2$, for considering up-regulation and down-regulation. Then, we merged two unified probabilities for each gene, one for up-regulation ($P\text{-up}$), and the other for down-regulation ($P\text{-down}$) using Fisher’s method (34). The test statistic (χ^2) following a chi-square distribution is shown below.

$$\chi^2 = -2 \sum_{i=1}^k \ln(p_i)$$

We used the ‘fisher.method’ function in the R package metaseqR (35) to compute the statistics and the corresponding P -values. Genes with only one significant P -value (either $P\text{-up} < 0.05$ or $P\text{-down} < 0.05$) were retained in the DEG list, and the directions of their dysregulations were determined by assessing which P -value was significant. Similarly, we also required an average $|\log_2(\text{fold change})| \geq 0.5$ based on those EXPs with significant differences in the individual tests. Based on the merged set of differentially expressed genes for each herb/ingredient, we performed separate GO and KEGG enrichment analyses for all, up and down-regulated genes, respectively. Furthermore, we adjusted up and down P -values separately to account for false discovery rates (FDRs) using the BH method (36).

Data-driven mapping of TCM herbs/ingredients with modern drugs

Once we established the gene expression profiles for TCM herbs/ingredients, we further evaluated the similarities between TCM herbs/ingredients and modern drugs by mapping HERB-EXP to CMap, which contains pharmacotranscriptomics datasets for thousands of well-annotated small molecules. We thus built the data-driven connections by first mapping the DEG list derived from each EXP to CMap and then merged the mapped results from all EXPs from a given herb or ingredient. Note that data from EXPs performed in *H. sapiens* or *M. musculus* were mapped and displayed separately.

For each EXP, the list of differentially expressed genes was submitted to the CMap website (<https://clue.io/query>) in batch query mode. Of note, a maximum of the top 300 genes in the DEG list were retained (5). It is noteworthy that the DEG genes from mouse EXPs were firstly converted to their human orthologs by using the R package homologene with the conversion table from NCBI (<ftp://ftp.ncbi.nih.gov/pub/HomoloGene/build68/>) (Supplementary Table

S2), 然后将每只小鼠 EXP 的人类基因提交进行 CMap 计算。我们查询的 CMap 数据集称为 Touchstone, 包含对 2837 种化合物在九种细胞系中进行扰动实验的数据。使用连接图 (connectivity map) 方法 (3), 评估输入 DEG 列表与所有这 2837 种扰动剂的基因表达谱之间的相似性, 并根据它们的连接图评分进行排序, τ , 排名从 -100 到 100 。 τ 为 $+90/-90$ 表示只有 10% 的随机扰动表现出比输入 DEG 列表更强的连接性, 而“加”或“减”分别表示正相关或负相关。然后, 我们将每种草药成分与多个实验 (EXP) 的连通性结果合并。我们遵循 CMap (3) 中的最大分位数统计方法, 将来自每次比较 (EXP) 的连通性图得分合并。每种草药成分的汇总连通性图得分根据以下公式计算:

$$\text{得分}_{h,p} = \begin{cases} Q_{\text{high}}(\text{得分}_{s,h,p}) & \text{如果 } |Q_{\text{high}}(\text{得分}_{s,h,p})| \geq |Q_{\text{low}}(\text{得分}_{s,h,p})| \\ Q_{\text{low}}(\text{得分}_{s,h,p}) & \text{其他 } w \text{ 情况} \end{cases}$$

最终分数 $_{h,p}$ 表示HERB中草药/成分h与CMap数据库中扰动剂p之间的整体相似度, 而分数 $_{e,h,p}$ 表示草药/成分h相关的每个EXP e与扰动剂p之间的个体相似度。该过程比较了 $\text{score}_{e,h,p}$ 的 Q_{high} 和 Q_{low} 分位数, 并保留绝对值较大的一个。 Q_{high} 和 Q_{low} 的分数被设置为0.67和0.33, 正如CMap所建议的。

手工整理的中药草药和成分参考数据

我们通过PubMed文本挖掘收集了2011年至2020年的中医相关论文。以草药和成分的名称及别名作为关键词, 要求关键词必须出现在标题或摘要中。为了缩小初始论文集合, 我们首先要求参考文献应发表在《期刊引证报告2019》中影响因子高于五的期刊上, 并且仅从中国科学院国家科学图书馆 (<http://www.fenqubiao.com/>) 分类系统下的生物医学相关类别中选择文章。然后, 我们将搜索范围限制在PubMed中未标记为综述论文的研究文章。第三, 我们只保留与纳入中医化学成分国家数据库的活性草药成分相关的论文 (见材料与方法第一节)。经过PubMed挖掘和分层筛选后, 我们获得了一个

招募了十名博士候选人来阅读、筛选并提取这套 17,886条文献中的信息, 高年级学生则进行二次验证。此工作最终从其中1966条文献中获得了可用的关于中药草药成分的基因靶点和相关疾病的信息, 这些文献被称为HERB-REF, 其中大多数都是低通量实验提供的证据。因此, 获得的数据不仅包含了有关…的信息

不仅关注草药/成分与靶标/疾病的关系, 还关注实验中使用的细胞系、动物模型或患者。此外, 我们根据NCBI命名法和GeneCards数据库(37)对新整理的中药靶标基因名称进行了标准化, 然后将其与广泛使用的现代化合物治疗靶标数据库TTD(17)进行了交叉参考。随后, 我们将疾病信息标准化为MeSH术语 (<https://www.nlm.nih.gov/mesh/meshhome.html>)和DisGeNet数据库(18), 并与HPO和疾病本体(20)数据库进行了交叉参考。此外, 我们在HERB-REF中整理了一些临床相关表型以方便用户使用。最后, 我们将DisGeNet(18)中靶标与疾病之间的关系转移至HERB。通过这种方法, HERB提供了一组基于数据的关联, 将中医药与现代药物联系起来, 建立在高质量手工整理参考文献的基础上。

HERB成分之间的其他关联

在 HERB-REF 之前, 像 SymMap (21)、HIT (15)、TCMSP (23) 和 TCMID (22) 这样的数据库也整理了成分-靶点关系。因此, 我们将参考文献和数据库挖掘所得的成分-靶点对合并在一起。其他关系, 例如成分-疾病、药材-靶点和药材-疾病, 通常通过组合两个或多个直接关系间接关联。例如, 成分与疾病之间的间接关系可以通过基因靶点作为中介获得 (补充图 S1A)。由于 HERB 包含最全面的药材和靶点列表, 我们使用 Fisher 精确检验重新计算了这三种关系的间接关联, 这称为统计推断, 并借鉴自 SymMap。以成分-疾病关系为例, 我们首先通过连接成分-靶点和靶点-疾病关系获得所有间接关联, 然后选择可靠的关联

HERB的实施

总之, HERB 提供关于草药、成分、它们的基因靶点、以现代术语描述的疾病以及相关的高通量实验数据和人工整理的参考文献的信息。HERB 提供便捷的网页界面, 供用户浏览、搜索、可视化和下载数据。HERB 可以免费访问, 网址为 <http://herb.ac.cn>, 无需用户注册。HERB 网站使用 Python-Flask、Nginx 和 React JavaScript 框架构建, 兼容大多数主流浏览器。HERB 的数据存储在 MySQL 数据库中。

S2), and then the human genes for each mouse EXP were submitted for CMap calculation. The CMap dataset we queried, called Touchstone, consisted of perturbation data for 2837 compounds that were tested in nine cell lines. Using the connectivity map method (3), the similarities between the input DEG list and the gene expression profiles for all these 2837 perturbagens were evaluated and ranked by their connectivity map scores, τ , ranked from -100 to 100 . A τ of $+90/-90$ indicates that only 10% of random perturbations showed stronger connectivity to the input DEG list, and ‘plus’ or ‘minus’ represents a positive or negative correlation, respectively. Then, we merged the connectivity results for each herb/ingredient with multiple EXPs. We followed the method of maximum quantile statistic in CMap (3) to merge the connectivity map scores derived from each comparison (EXP). The summarized connectivity map score for each herb/ingredient was calculated according to the following formula,

$$\text{Score}_{h,p} = \begin{cases} Q_{\text{high}}(\text{score}_{e,h,p}) & \text{if } |Q_{\text{high}}(\text{score}_{e,h,p})| \geq |Q_{\text{low}}(\text{score}_{e,h,p})| \\ Q_{\text{low}}(\text{score}_{e,h,p}) & \text{otherwise} \end{cases}$$

where the final $\text{score}_{h,p}$ stands for the overall similarity measure between a herb/ingredient h in HERB and a perturbagen p in the CMap database, and the $\text{score}_{e,h,p}$ indicates the individual similarity measure between each EXP e related to the herb/ingredient h and the perturbagen p . This procedure compares the Q_{high} and Q_{low} quantiles of $\text{score}_{e,h,p}$ and retains whichever is of higher absolute magnitude. The scores of Q_{high} and Q_{low} were set to be 0.67 and 0.33 respectively, as suggested by CMap.

Manually curated reference data for TCM herbs and ingredients

We collected TCM-related papers from 2011 to 2020 through PubMed text mining. The names and aliases of herbs and ingredients were used as key words, and the key words were required to be present in the title or abstract. In order to narrow down the initial paper set, we first required that references should be published in journals with an impact factor above five according to the Journal Citation Report 2019, and we only selected from articles classified in the biomedical-related categories according to the classification system of the National Science Library, Chinese Academy of Sciences (<http://www.fenqubiao.com/>). Then, we restricted our search to research articles that were not labeled as review papers in PubMed. Third, we only kept those papers that were related to active herbal ingredients that were included in the National Database for Chemical Composition in TCM (see the first section in Materials and Methods). After PubMed mining and hierarchical filtering, we obtained a list of 17 886 references for manual curation.

Ten Ph.D. candidates were recruited to read, select, and extract information from this set of 17 886 references, with senior students performing a secondary verification. This effort resulted in a set of usable information on gene targets and diseases related to TCM herbs/ingredients from 1966 of these references, called HERB-REF, most of which had evidence from low-throughput experiments. Thus, the resulting data included information not only on

herb/ingredient and target/disease relationships, but also on the cell line, animal model, or patients used in the experiments. We further standardized gene names for the newly curated TCM targets according to the NCBI nomenclature and GeneCards database (37), and then cross-referenced them with the TTD database (17), a widely used therapeutic target database for modern compounds. Then, we standardized the disease information to MeSH terms (<https://www.nlm.nih.gov/mesh/meshhome.html>) and the DisGeNet database (18), and cross-referenced them with the HPO and Disease Ontology (20) databases. Further, we curated a number of clinically relevant phenotypes in HERB-REF for users’ convenience. Finally, we transferred the relationships between targets and diseases from DisGeNet (18) to HERB. As a result of this approach, HERB offers a set of data-based connections linking TCM and modern drugs, built on basis of high-quality manual curation of references.

Other associations among HERB components

Prior to HERB-REF, databases like SymMap (21), HIT (15), TCMSP (23) and TCMID (22) have also curated ingredient-target relationships. Therefore, we merged together the ingredient–target pairs from reference and database mining. Other relationships, e.g. ingredient–disease, herb–target and herb–disease, were generally indirectly linked by combining two or more direct relationships. For example, the indirect relationship between ingredient and disease can be obtained using the gene targets as a middle component (Supplementary Figure S1A). As HERB contains the most comprehensive list of herbs and targets, we calculated the indirect associations for these three relationships again using Fisher’s exact test, which is called statistical inference and adopted from SymMap. Taking the ingredient-disease relationship as an example, we first acquired all indirect associations by linking ingredient-target and target-disease relationships together and then selected reliable associations (FDR < 0.01) from them using Fisher’s exact test followed by multiple test corrections using the BH method (36). This strategy was also used to infer the herb–target and herb–disease relationships by using ingredient and target as the middle components, respectively (Supplementary Figures S1B, C). Finally, we combined the associated pairs from statistical inference and reference mining together.

Implementation of HERB

In summary, HERB provides information about herbs, ingredients, their gene targets, diseases in modern terms, and the relevant high-throughput experimental data and manually curated references. HERB provides a convenient web interface for users to browse, search, visualize, and download data. HERB is freely accessible at <http://herb.ac.cn> without a need for user registration. The HERB website was built using the Python-Flask, Nginx and React JavaScript frameworks and is compatible with most major browsers. The HERB data are stored in a MySQL database.

表 1. HERB 数据汇总概况

组件	数量	数据来源
草药	7263	整合自SymMap、TCMID、TCMSP和TCM-ID数据库
成分	49 258	整合自SymMap、TCMID、TCMSP和TCM-ID数据库
实验	1037	从 NCBI GEO 数据库下载、人工整理并自动分析
参考文献	1966	从 NCBI PubMed 数据库下载、手动整理并提取信息
目标	12,933	部分来自PubMed参考文献的人工整理，部分来自以前的数据库包括SymMap、HIT、TCMSP和TCMID
疾病	28 212	部分来自PubMed参考文献的手动整理，部分来自DisGeNet数据库

结果

HERB中的数据内容

HERB 包含一个全面的中药草药（7263种）和成分（49,258种）列表，通过整合多个中医药数据库而成。HERB 是通过重新分析来自1037个高通量实验的6164个基因表达谱以及收集的1966个与中医相关的文献创建的，从中整理出了473种药材/成分对应的1241个基因靶点和494种现代疾病，以及709个临床相关表型。此外，HERB 还包括通过数据库挖掘和统计推断获得的靶点和疾病信息，最终得出与药材和成分相关的总共12,933个靶点和28,212种疾病。这些数据统计显示在表1中。我们还提供了六种药材、成分、靶点和疾病之间的成对关系的统计，这些关系通过文献挖掘、数据库挖掘和/或通过统计推断的间接关联获得（补充表S3）。HERB 包含两个重要的新特性

在HERB中分析的高通量数据

通过使用草药和成分作为关键词进行GEO挖掘，我们收集了472个高通量GEO数据集，包含6164个GEO样本。每个由特定实验室完成的GEO数据集都有一个唯一的GSE编号。每个在特定生物生理条件下进行的GEO样本也都有一个唯一的GSM编号。我们通过定义HERB实验（EXP）来手动获取以草药/成分为中心的数据分类，将其定义为一个GEO数据集中与某种草药/成分相关的一组对照和处理GEO样本。这导致HERB中共有1037个EXP，其中83.9%为微阵列数据，其余16.1%为RNA-seq数据；HERB-EXPs中有10.4%与草药相关，89.6%与成分相关（图1A-1）。HERB-EXPs中对照组和处理组样本的平均生物重复数分别为4.9和3.7。值得注意的是，83.2%的对照样本和74.1%的处理样本至少有三个重复样本。

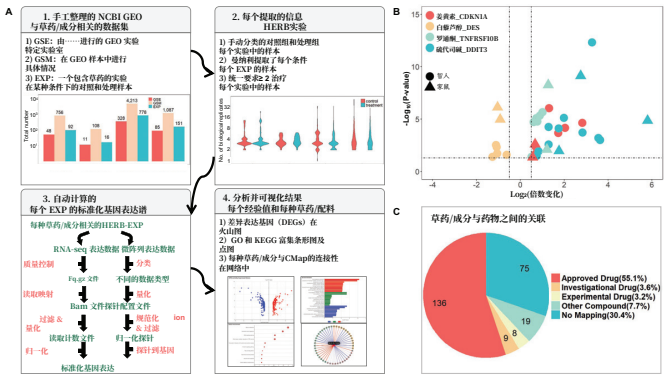


图1. HERB-EXP的构建与表征。(A) HERB-EXP数据处理框架的示意图，分为四个连续步骤：1. 从GEO实验（GSE）和样本（GSM）下载数据，提取每种草药/成分的HERB实验（EXP），并用柱状图显示每种数据类型的GSE/、GSM/、EXP数量。2. 对每个EXP，用小提琴图展示对照组和处理组样本的生物重复数量。3. 首先根据不同的分析流程自动分析每个EXP的RNA-seq和微阵列数据，然后将多个EXP的结果合并至每种草药/成分。4. 可视化每个EXP的差异表达分析、GO和KEGG富集或CMap连通性结果，然后分别对每种草药/成分进行火山图、柱状图、点图或网络图展示。(B) 四种已知靶点（CDKN1A、DES、TNFRSF10B和DDIT3）在四种中药成分（姜黄素、r

（图 1A-2）。足够数量的重复实验有助于对这些高通量数据进行后续分析。然后，我们建立了自动化流程来分析每个实验（EXP）的微阵列和RNA-seq数据（图1A-3）。为了处理RNA-seq数据，我们以原始数据为输入，将过滤后的序列比对到基因组，并获得每个基因的归一化读数。对于微阵列原始数据，我们首先将数据处理为探针表达谱，然后将其转换为归一化的基因表达数据。如果没有原始数据，则跳过前一步。基于每个实验中对照样本和处理样本的归一化基因表达矩阵，我们鉴定了每个实验的差异表达基因（DEGs），然后将结果合并为每种草药成分的排序DEG列表。我们通过火山图（图1A-4）可视化了草药及其成分的DEG列表。基于DEG列表，我们可视化了富集的GO术语。

Table 1. Overview of the data curated in HERB

Components	Amount	Data source
Herbs	7263	Integrated from SymMap, TCMID, TCMSP and TCM-ID databases
Ingredients	49 258	Integrated from SymMap, TCMID, TCMSP and TCM-ID databases
Experiments	1037	Downloaded, manually curated, and automatically analyzed from the NCBI GEO database
References	1966	Downloaded, manually curated, and extracted information from the NCBI PubMed database
Targets	12 933	Partially from the manual curation from PubMed references and others from previous databases including SymMap, HIT, TCMSP, and TCMID
Diseases	28 212	Partially from the manual curation from PubMed references and others from the DisGetNet database

RESULTS

Data contents in HERB

HERB contains a comprehensive list of TCM herbs (7263) and ingredients (49 258) by integrating multiple TCM databases. HERB was created through reanalysis of 6164 gene expression profiles from 1037 high-throughput experiments of herbs/ingredients and a collection of 1966 TCM-related references, from which 1241 gene targets and 494 modern diseases for 473 herbs/ingredients, as well as 709 clinical-relevant phenotypes were curated. Further, HERB includes information on targets and diseases gathered by database mining and statistical inference, resulting in a final total of 12 933 targets and 28 212 diseases related to herbs and ingredients. These data statistics are shown in Table 1. We also provided the statistics of six pairwise relationships among herbs, ingredients, targets, and diseases that were gathered from reference mining, database mining, and/or indirect association through statistical inference (Supplementary Table S3). HERB contains two important novel features lacking from previous TCM databases. First, HERB presents a comprehensive pharmacotranscriptomics data set for TCM herbs and ingredients, allowing mapping to modern drugs with data in CMap. Second, HERB provides manually curated data of targets and diseases mined from recent literature, providing high-confidence information for ranking TCM herbs and ingredients as promising candidates for therapeutic development.

High-throughput data analyzed in HERB

Through GEO mining using herbs and ingredients as key words, we gathered 472 high-throughput GEO datasets containing 6164 GEO samples. Each GEO dataset performed by a specific lab had a unique GSE number. Each GEO sample conducted in a particular biological/physiological condition also had a unique GSM number. We manually obtained the herb/ingredient-centric classification of these data by defining a HERB experiment (EXP), as a set of control and treatment GEO samples related to a herb/ingredient in one GEO dataset. This resulted in 1037 EXPs in HERB, with 83.9% of them consisting of microarray data and the remaining 16.1% as RNA-seq data; 10.4% of HERB-EXPs are related to herbs and 89.6% of them are related to ingredients (Figure 1A-1). The average number of biological replicates for control and treatment samples in HERB-EXPs were 4.0 ± 2.9 and 3.7 ± 2.6 respectively. Of note, 83.2% and 74.1% of control and treatment samples had at least three replicates, respectively

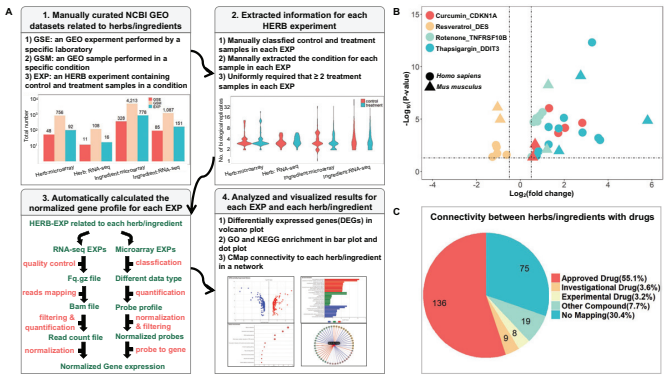


Figure 1. Construction and characterization of HERB-EXP. (A) A schematic diagram of the data processing framework in HERB-EXP in four consecutive steps: 1. download data from GEO experiments (GSE) and samples (GSM), extract HERB experiments (EXP) for each herb/ingredient, and show the number of GSE/GSM/EXPs for each data type in bar plots. 2. For each EXP, illustrate the number of biological replicates for control and treatment samples in violin plots. 3. Automatically analyze RNA-seq and microarray data according to different pipelines for each EXP first, and then merge results from multiple EXPs together for each herb/ingredient. 4. Visualize the results of differential expression analysis, GO and KEGG enrichment, or CMap connectivity for each EXP, and then for each herb/ingredient, in volcano plot, bar plot, dot plot or network respectively. (B) The differential expression of four known targets (CDKN1A, DES, TNFRSF10B and DDIT3) for four TCM ingredients (curcumin, resveratrol, rotenone, and thapsigargin) in multiple EXPs related to these ingredients are shown in volcano plots. The X-axis shows the expression change compared to control samples and the Y-axis shows statistical significance. The four colors correspond to four target-ingredient pairs, and the two shapes indicate the species information for each EXP (mouse/human). (C) Classification of all herbs/ingredients with HERB-EXP data, according to their mapped compounds in CMap.

(Figure 1A-2). Sufficient numbers of replicates facilitate downstream analysis of these high-throughput data. Then, we built automatic pipelines to analyze the microarray and RNA-seq data of each individual EXP (Figure 1A-3). To process the RNA-seq data, we used raw data as an input, mapped the filtered reads to the genome, and got the normalized read count for each gene. For microarray raw data, we first processed the data into a probe profile and then converted it into normalized gene expression data. If no raw data were available, the former step was skipped. Based on the normalized gene expression matrix for control and treatment samples in each EXP, we identified DEGs for each EXP and then merged the results into a ranked DEG list for each herb/ingredient. We visualized the DEG list for herbs and ingredients by volcano plot (Figure 1A-4). Based on the DEG list, we visualized the enriched GO terms

并在柱状图和点图中显示所有、上调和下调基因的KEGG通路(图1A-4)。此外, 我们查询了包含2837种现代药物的CMap数据的药物转录组学数据集, 同时使用每种草药成分的差异表达基因(DEG)列表作为查询。草药成分与其可映射现代药物之间的关系通过交互式网络可视化(图1A-4)。

总而言之, 在EXP水平分析中平均有1801个差异表达基因(DEGs)(补充图S2A), 而在合并多个EXP结果后, 草药成分平均有2864个DEGs(补充图S2B)。对于GO术语, EXP水平和草药成分水平的平均数量分别为1083和1260(补充图S3)。对于KEGG通路, 这两个数字分别为44和50(补充图S4)。我们还展示了上调和下调基因的GO及KEGG富集结果(补充图S3和S4)。接下来, 我们选取了所有在已有文献中描述的与中药成分相关的DEGs, 并探索了它们在HERB实验中的表达变化。在325个这样的成分-靶点对中, 有208个(64%)显示出一致的差异表达模式(补充表S4)。其他不一致可能是由转录后调控问题引起的, 有相关证据支持。

接下来, 我们通过将HERB-EXP数据与CMap数据进行连通性映射, 将中药及其成分与现代药物化合物连接起来。总共有20种中药和152种成分可以通过绝对连通性评分高于95的阈值映射到978种CMap化合物。通过将CMap中记录的ID(InChiKey或PubChem ID)与DrugBank连接, 可以将978种CMap化合物中的383种的临床状态分类为批准药物、在研药物、实验药物或其他化合物。因此, 我们进一步根据中药成分所映射的化合物类型对其进行分类。在HERB中拥有高通量数据的247种中药成分中, 有136种(55.1%)能够映射到批准药物。另外有9种(3.6%)和8种(3.2%)中药成分分别能映射到处于临床试验的在研药物和处于临床前研究的实验药物(图1C)。综合起来

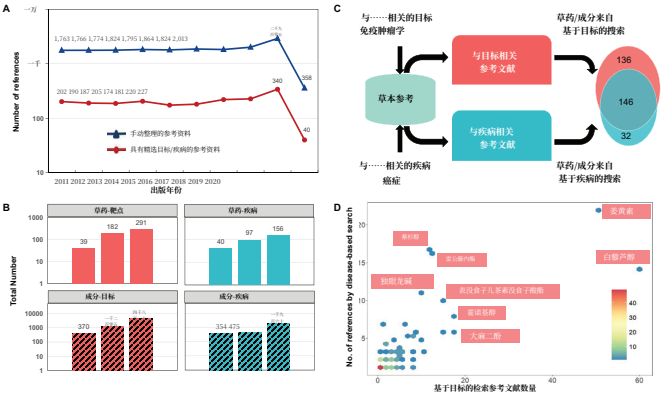


图2. HERB-REF概述。(A) 各年份发表的与草药/成分相关参考文献数量统计。蓝线表示手工整理的总参考文献数, 红线表示具有整理过的靶点/疾病的参考文献数。(B) HERB-REF中整理的草药/成分、靶点/疾病及其关联数量以柱状图显示, 四个子图分别展示四种类型的关联: 草药-靶点、草药-疾病、成分-靶点和成分-疾病。(C) 使用HERB-REF数据集搜索并排序可能用于与免疫肿瘤学相关靶点相关的癌症相关疾病的潜在草药或成分。靶点搜索和/或基于疾病的搜索获得的草药/成分数量显示在维恩图中。(D) 通过靶点搜索和疾病搜索均支持的选定草药/成分以点图形式展示, X轴和Y轴显示数量

基因以一对一的方式进行了转换。平均而言, 这些EXP中65.3%的小鼠DEG基因可以被映射(补充图S5)。因此, 基于小鼠数据的CMap映射结果可能不完整。所以小鼠数据仅作为辅助信息提供。

HERB中整理的参考文献

我们通过PubMed文本挖掘收集了2011年至2020年间与中医相关的论文, 获得了17,886个参考文献, 供分层过滤后手动整理(见图2A, 蓝线)。过去三年中, 与中药成分相关的发表论文数量大幅增加, 2019年发表的相关论文有2905篇位居首位(部分数据为2020年)。在手动阅读、筛选和提取最初17886条参考文献后, 1966条文献被单独检索, 我们从中收集了大量高质量的中药成分基因靶点和现代疾病数据。最终一组未筛选目标/疾病参考文献(图2A, 红线)与最初手工管理的参考文献集(图2A, 蓝线)非常相似, 表明10名博士候选人手工筛选过程是无偏的。仅选中了10.9%的初始参考文献, b

and KEGG pathways in bar plots and dot plots, for all, up and down-regulated genes (Figure 1A-4), respectively. Moreover, we queried the CMap data containing pharmacotranscriptomics datasets for 2837 modern drugs while using the DEG lists of each herb/ingredient as the query. The connections between the TCM herb/ingredient and their map-able modern drugs were visualized with a interactive network (Figure 1A-4).

In summary, there were an average of 1801 DEGs upon EXP-level analysis (Supplementary Figure S2A) and 2864 DEGs, on average, for herbs/ingredients after merging the results from multiple EXPs (Supplementary Figure S2B). For GO terms, the average numbers were 1083 and 1260 for the EXP-level and herb/ingredient-level results (Supplementary Figure S3). For KEGG pathways, the two numbers were 44 and 50, respectively (Supplementary Figure S4). We also illustrated the GO/KEGG enrichment results for up-regulated and down-regulated genes (Supplementary Figures S3 and S4). Next, we selected all DEGs related to TCM ingredients that were described in previously published papers and explored their expression changes in the HERB experiments. Across 325 such ingredient-target pairs, 208 of them (64%) showed consistent patterns of differential expression (Supplementary Table S4). Other inconsistencies may be caused by post-transcriptional regulation issues, as evidences from references are mainly based on protein expression levels. We also showed 4 distinguished ingredient-gene pairs in Figure 1B. The direction of regulation for them, including the curcumin-CDKN1A (38), resveratrol-DES (39), rotenone-TNFRSF10B (40) and thapsigargin-DDIT3 (41), were the same as identified in their corresponding references, and cross-validated by both human and mouse data that are merged separately. Thus, they are most likely to be the *bona fide* targets that are highly related to the downstream pathways affected by these ingredients. Next, we connected TCM herbs and ingredients to modern pharmaceutical compounds using connectivity mapping between HERB-EXP data and CMap data. In total, 20 herbs and 152 ingredients could be mapped to 978 CMap compounds using a cutoff of absolute connectivity score above 95. The clinical status for 383 out of 978 CMap compounds could be classified as approved drugs, investigational drugs, experimental drugs, or other compounds, by connecting their IDs (InChiKey or PubChem ID) documented in CMap to DrugBank. Thus, we further classified herbs/ingredients according to the type of compounds to which they were mapped. Out of 247 herbs/ingredient with high-throughput data in HERB, there were 136 (55.1%) herbs and ingredients that could be mapped to approved drugs. Another 9 (3.6%) and 8 (3.2%) herbs/ingredients could be mapped to investigational drugs that were in clinical trials and experimental drugs that were under preclinical research, respectively (Figure 1C). Taken together, the gene expression patterns for >60% of herbs/ingredients were similar or antagonistic to modern drugs. These findings illustrate the potential for other TCM herbs/ingredients to be investigated in pharmaceutical development. We have provided a detailed list of the mapped drugs for the TCM herbs/ingredients in Supplementary Table S5. Note that for mouse EXP, a total of 15 123 mouse DEG genes can be converted to their human orthologs, from which 15 060 mouse

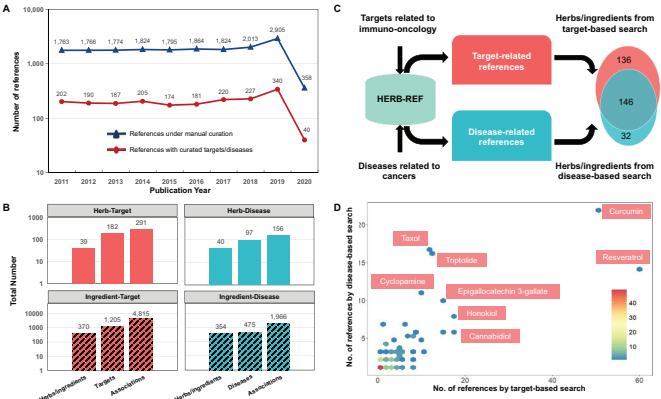


Figure 2. Overview of HERB-REF. (A) Statistics on the number of herb/ingredient-related references published in each year. The blue line represents the total references from manual curation, and the red line represents the references with curated targets/diseases. (B) The number of curated herbs/ingredients, targets/diseases, and their associations from HERB-REF are shown in bar plots, with the four facets showing four types of associations: herb-target, herb-disease, ingredient-target, and ingredient-disease. (C) HERB-REF datasets were used to search and rank promising herbs or ingredients for cancer-related diseases that could be connected to immuno-oncology related targets. The number of herbs/ingredients obtained by target-based search and/or disease-based search was shown in a Venn plot. (D) The selected herbs/ingredients that were supported by both target-based and disease-based search were illustrated in a dot plot, with the X- and Y-axis showing the number of references from each search, respectively. The density of dots representing the number of dots at the same point are represented in color scales.

genes were converted in a one-versus-one manner. On average, 65.3% of mouse DEG genes in these EXPs can be mapped (Supplementary Figure S5). As a result, the CMap mapping results based on mouse data maybe incomplete. So the mouse data was just provided as an auxiliary information.

References curated in HERB

We collected TCM-related papers from 2011 to 2020 by PubMed text mining and obtained a list of 17 886 references for manual curation after hierarchical filtering (Figure 2A, blue line). The number of published papers related to TCM herbs/ingredients has increased dramatically in the past three years, with 2905 relevant papers published in 2019 at the top (with partial data for 2020). After manually reading, selecting, and extracting information from the initial 17 886 references, 1966 references were selected, from which we collected a large amount of high-quality data on gene targets and modern diseases for TCM herbs/ingredients. The pattern of the final set of references with curated targets/diseases (Figure 2A, red line) was quite similar with the initial set of references under manual curation (Figure 2A, blue line), suggesting that the process of manual curation by 10 Ph.D. candidates was unbiased. Only 10.9% of the initial set of references were selected, because we carefully filtered out a large number of references without relevant or reliable information. Finally, we manually extracted 1241 gene targets of 39 herbs and 370 ingredients and manually connected 494 modern diseases with 40 herbs and 354 active ingredients (Figure 2B). The num-

新整理的靶点数量与 HIT 2011 (1301 个) 相似, 并且多于 TCMID 2012 (680 个), 表明近年来持续有大量中医药研究集中于分子和机制研究。这套新的整理疾病信息包括直接的实验数据, 因此其外部有效性可能高于此前通过基因靶点间接将草药成分与疾病联系起来的方法。此外, 还整理了 709 个临床相关的表型, 尽管它们本身不是疾病, 但为了用户的方便也进行了展示。

接下来, 我们根据新整理的 HERB-REF 数据, 对某种疾病的候选靶点进行排序, 筛选出具有潜力的草药或成分。例如, 由于中医被证明在调节癌症患者免疫方面具有临床益处 (42), 我们使用与免疫肿瘤学相关的靶点和疾病作为输入查询 HERB-REF。我们首先根据外部已发表的论文 (43–45) 收集了 400 个与免疫肿瘤学相关的靶基因列表。然后, 我们使用关键词 “neoplasm”、“tumor”、“cancer” 或 “carcinoma” 在 HERB-REF 的 1,966 条参考文献中搜索了 53 种与癌症相关的疾病。我们将两个列表与靶点相关或疾病相关的草药或成分进行交集 (图 2C)。结果得到了 282 种在免疫肿瘤学相关靶点列表中有靶点支持并被 HERB-REF 中 745 篇文献佐证的草药 / 成分, 同时获得了 178 种可与免疫肿瘤学相关疾病联系的草药/成分。

我们进一步在图2D中可视化了与免疫肿瘤学相关的 146种草药/成分。大多数这些草药/成分仅得到少量文献支持, 无论是基于靶点搜索还是基于疾病搜索, 这一点可从左下角高密度的点位看出。右上角的点表示有大量参考文献支持的高置信度估计。例如, 由姜黄植物 (*Curcuma longa*) 产生的活性成分姜黄素能够抑制 COP9 信号体 5 (CSN5) 并降低癌细胞中 PD-L1 的表达, 这是众所周知的癌症免疫治疗靶点。因此, 姜黄素可能是癌症联合疗法的有前景的候选药物。在我们的 HERB-REF 搜索中, 姜黄素与免疫肿瘤学的关联得到了 65 篇文献的支持, 其中 51 篇为基于靶点的文献, 22 篇为基于疾病的文献, 8 篇同时来自两种搜索, 显示了 HERB 工具的强大作用。

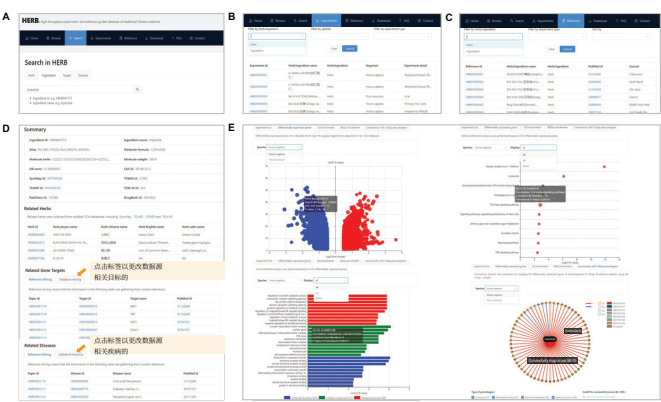


图3. HERB 搜索示意图。(A) HERB 的搜索页面显示了 HERB 的四个主要组成部分, 包括草药、成分、靶点和疾病。(B) HERB 的实验页面显示了所有与草药成分相关并由 HERB 分析的高通量数据。(C) HERB 的参考页面显示了所有已发表的文献, 其中包含经过整理的草药成分靶点和疾病信息。(D) 示例成分雷公藤内酯的详细信息页面。首先显示的是摘要面板, 展示该成分的相关描述性信息。随后显示的面板分别展示与该成分相关的草药、基因靶点和疾病。用户可以点击每个面板中的标签来浏览来自不同数据源的信息。(E) 雷公藤内酯的后续详细页面。使用四个图形来可视化与雷公藤内酯相关的高通量数据。例如, 差异表达基因在火山图中显示 (左上)。GO 富集条目和

除了参考资料挖掘之外, 我们还整合了先前中医药数据库 (包括 SymMap、HIT、TCMSP 和 TCMID) 中与 7781 种成分和 12 838 个基因靶点相关的 157 722 对成分-靶点配对。我们统计推断了成分-疾病、药材-靶点和药材-疾病之间的间接关系 (补充表 S3)。基于不同阈值, 每种成分及其关系的数量如补充图 S6 所示。因此, 我们最终将 12 933 个靶点和 28 212 种疾病与 7263 种药材以及 49 258 种成分关联起来, 从而获得 HERB, 这是迄今为止最全面的中医药数据库。

使用 HERB 数据库

HERB 可以通过 <http://herb.ac.cn> 免费访问。它为用户提供浏览和搜索页面, 以便浏览草药、成分、靶点和疾病。每个组成部分的搜索菜单中或下方显示了可搜索关键词的示例和类型 (图 3A)。对于 HERB 中添加的两个新功能——实验和文献, 我们提供了额外的浏览和搜索页面。在实验搜索页面 (图 3B) 中, 用户可以点击实验 ID 查看每个实验 (EXP) 的详细信息页面。关于每个 EXP 的描述性信息包括实验对象、实验条件、给药方法、处理组和对照组的样本信息等。这些描述可以在每个 EXP 的搜索页面和详细页面中找到。EXP 的分析结果

ber of newly curated targets was similar to that of HIT 2011 (1301), and larger than TCMID 2012 (680), demonstrating that a continuous number of recent TCM studies have focused on molecular and mechanistic studies. This new set of curated disease information included direct experimental data, and is thus likely of external validity than prior approaches that indirectly linked herbs/ingredients to diseases using gene targets as intermediates. Further, 709 clinically relevant phenotypes, although not diseases themselves, were curated and presented for users’ convenience.

Next, we ranked herbs or ingredients that were promising for a particular disease with a set of candidate targets, based on the newly curated HERB-REF data. For example, as TCM has been shown to have clinical benefits for regulating the immunity of cancer patients (42), we used the targets and diseases related to immuno-oncology as the input to query HERB-REF. We first collected a list of 400 target genes related to immuno-oncology, according to external published papers (43–45). Then, we searched the 1,966 references in HERB-REF for 53 cancer-related diseases using the key words ‘neoplasm’, ‘tumor’, ‘cancer’ or ‘carcinoma’. We intersected the two lists with the target-related or disease-related herbs or ingredients (Figure 2C). This resulted in 282 herbs/ingredients that had targets in the immuno-oncology related list and were supported by 745 of the references in HERB-REF, and obtained 178 herbs/ingredients that could be connected to immuno-oncology related diseases and supported by 396 references. When we required both target and disease relationships, we were left with a final list of 9 herbs and 137 ingredients that were highly related to immuno-oncology. The list of 146 herbs/ingredients is provided in Supplementary Table S6.

We further visualized the 146 herbs/ingredients related to immuno-oncology in Figure 2D. Most of these herbs/ingredients were supported by a small number of references from either target-based search or disease-based search, as indicated by the high density of dots in the bottom-left corner. The dots in the upper-right corner represent higher confidence estimates with large numbers of supporting references. For example, curcumin, an active ingredient produced by the *Curcuma longa* plant, inhibits the COP9 signalosome 5 (CSN5) and diminishes the expression of PD-L1 in cancer cells, which is a well-known target for cancer immune therapies (46). As a result, curcumin may represent a promising candidate for combination therapy in cancer. In our HERB-REF search, the connection of curcumin to immuno-oncology was supported by 65 references, with 51 target-based references, 22 disease-based references, and 8 references from both searches, demonstrating the power of HERB-REF for searching and ranking TCM solutions for a given disease. Cannabidiol, a phytocannabinoid from cannabis plants, has long been shown to have anti-tumor activity (47). In both cell lines and an animal model of triple-negative breast cancer, cannabidiol significantly inhibits epidermal growth factor (EGF)-induced proliferation and inhibits the recruitment of tumor-associated macrophages (47). In HERB-REF, cannabidiol was supported by 6/17/4 references from only target-based search, disease-based search, and both searches, respectively.

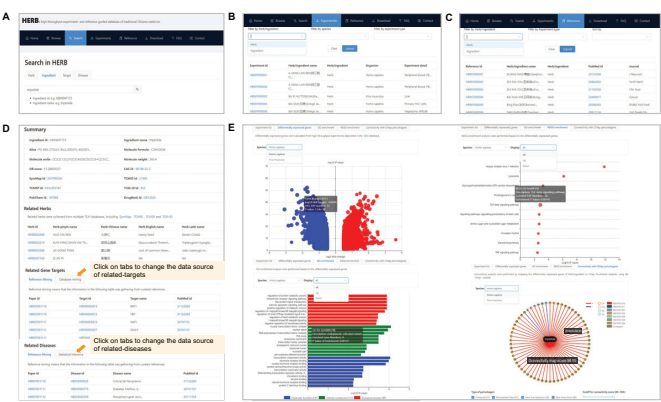


Figure 3. An illustration of an HERB search. (A) The search page of HERB shows the main 4 components of HERB, including herbs, ingredients, targets, and diseases. (B) The experiment page of HERB shows all high-throughput data related to herbs/ingredients and analyzed by HERB. (C) The reference page of HERB shows all published references with curated targets/diseases information for herbs/ingredients. (D) The details page for an example ingredient, triptolide. The summary panel comes first, and shows relevant descriptive information for this ingredient. The following panels show the herbs, gene targets, and diseases related to this ingredient, respectively. Users can click on the tabs in each panel to navigate information from different data sources. (E) The subsequent detailed page for triptolide. Four figures were used to visualize the high-throughput data related to triptolide. For example, the DEGs are shown in a volcano plot (upper-left). The enriched GO terms and KEGG pathways are shown in bar plot and dot plot (lower-left and upper-right). The connectivity between triptolide and its related compounds in CMap are shown in a network. Note that all figures are implemented in an interactive way where users can see the details of each dot/bar/node/edge upon mousing over them.

Aside from reference mining, we merged 157 722 ingredient-target pairs related to 7781 ingredients and 12 838 gene targets from previous TCM databases, including SymMap, HIT, TCMSP and TCMID. We statistically inferred the indirect relationships between ingredient-disease, herb-target, and herb-disease (Supplementary Table S3). The number of each component and their relationships, based on different cutoffs, are shown in Supplementary Figure S6. As a result, we ultimately linked 12 933 targets and 28 212 diseases to 7263 herbs and 49 258 ingredients to obtain HERB, the most comprehensive database for TCM available to date.

Using the HERB database

HERB is freely accessible at <http://herb.ac.cn>. It provides browse and search pages for users to navigate herbs, ingredients, targets, and diseases. Examples and the types of searchable keywords are shown in or below the search menu for each component (Figure 3A). For the two novel features added in HERB, experiments and references, we provide additional browsing and searching pages. In the experiment search page (Figure 3B), users can click on an experiment ID to view the details page for each EXP. Descriptive information about each EXP includes experimental subjects, experimental conditions, administration methods, sample information for treatment groups and control groups, etc. These descriptions can be found in both the search page and detailed page of each EXP. The analyzed results of EXP-

在实验的详细页面中, 以与草药成分相同的格式显示高通量数据, 如下一段所述。在参考文献搜索页面 (图3C) 中, 用户可以点击参考文献ID来查看每条参考文献的详细页面。在浏览页面和详细页面中显示的参考文献详情包括期刊名称、出版日期、实验对象、实验类型和 PubMed ID。在详细页面中, 我们展示了从每条对应的参考文献中手动整理的草药成分与靶点疾病之间的配对关系。此外, 我们还使用词云可视化了每条参考文献的摘要。

图 3D 和 E 显示了示例成分雷公藤内酯的详细页面。第一页包括关于雷公藤内酯及其相关草药、基因靶点和疾病的描述性信息 (图 3D)。汇总信息包括成分名称、别名、化学式、分子 SMILES、分子量以及来自其他数据库的外部 ID。雷公藤内酯的相关草药来自多个中医药数据库整合。雷公藤内酯的相关靶点和疾病来源于文献挖掘、数据库挖掘或统计推断, 每种信息都在不同的子页面下单独显示, 每个表格均显示数据来源说明。其中, 来自文献挖掘的记录 (即人工整理) 提供最高可信度的数据, 并可在 HERB 的参考页面中交叉引用。来自数据库挖掘的记录质量也较高, 并提供原始数据库名称及链接。其意义在于

HERB分析的高通量数据以表格和图表的形式可视化。用户可以根据所有表格中的每一列对记录进行排序和筛选。我们在网站中引入了四个图形, 以更好地可视化这些表格中的数据, 每个图形代表高通量数据的一个重要方面 (图3E)。请注意, 人类和小鼠的数据是分开显示的, 用户可以通过下拉菜单在这两个数据集之间切换。例如, 左上方显示的第一个火山图可视化了与雷公藤相关的差异表达基因 (DEGs), 这些数据已从多个雷公藤实验 (EXP) 中合并。图中的一个点代表一个基因, 颜色表示其表达异常的方向 (上调或下调), 点的大小与支持该基因为DEG的实验数量成正比。左下方的条形图和右上方的点图分别显示了富集的GO术语和KEGG通路。底层表格中的每个术语都可以被链接

它周围是四种不同颜色的相关化合物, 这些颜色表示 CMap 干预的不同数据类型, 包括化合物 (CP)、干预类别 (PCL)、基因过表达 (OE)、基因敲低 (KD)。此外, 我们在每个周围节点中嵌入了饼图, 以显示支持其映射到雷公藤内酯的具体实验 (EXP)。雷公藤内酯与其映射化合物之间的边用两种颜色显示, 红色表示基因表达谱之间的正相关, 蓝色表示负相关。边的宽度与连通性评分的绝对值成正比。默认情况下, 我们可视化了绝对连通性评分为 >95 的映射结果。用户可以自由选择不同的阈值和 CMap 数据类型的子集, 以专注于感兴趣的特定数据。同时, 用户可以使用左上角的放大 / 缩小按钮来控制连通性图。

讨论

转录组分析可以对由于化合物干扰所引起的转录反应和调控变化进行定量测量, 使其成为发现药物靶点、评估药物疗效以及揭示可能副作用的有力方法。近年来, 来自实验室或临床的中药及其成分的高通量数据迅速积累, 但迄今为止尚未有良好的组织系统。此外, 过去十年中新发布的中药相关文献尚未被整理。因此, 在本研究中, 我们重新分析了所有可获得的中药成分微阵列和 RNA-seq数据, 并从最近发布的中药文献中整理了高置信度的靶点和疾病信息。结合数据库挖掘和统计推断, 我们最终建立了HERB, 这是迄今为止最全面的中药数据库。

HERB数据库的新颖之处包括: (i) HERB通过重新分析所有可用的中医高通量实验数据, 提供了一个全面且统一的中医药药物转录组学数据库。使用HERB, 研究人员和药物开发者可以查看原始数据以及基于数据的中医药成分与现代化合物之间的映射结果, 从而轻松探索中药成分的潜在作用机制, 并识别新的潜在有效疗法。(ii) HERB基于过去十年新发表文献的人工整理, 提供与中药成分相关的高可信靶点和疾病信息, 弥补了自 HIT(2011)和TCMID(2012)创建以来的巨大差距。利用 HERB中新整理的参考文献, 用户可以轻松搜索并排序针对某疾病的一组候选靶点的有前景的中药或成分。我们相信, 这种直观的工作流程将有助于研究人员

level, high-throughput data are shown in the details page of experiments in the same format as herbs/ingredients, which are described in the following paragraph. In the reference search page (Figure 3C), users can click on the reference ID to view the detail page for each REF. Details of references shown in the browse and details page include the journal name, publication date, experimental subjects, experimental type, and PubMed ID. In the detailed page, we show the pairing relationships between herbs/ingredients and targets/diseases that were manually curated from each corresponding reference. Furthermore, we visualized the abstract for each reference by using a word cloud.

The detail page for an example ingredient, triptolide, is shown in Figure 3D and E. The first section includes descriptive information about triptolide and its related herbs, gene targets, and diseases (Figure 3D). The summary information includes the ingredient name, alias, formula, molecule SMILES, molecular weight, and external IDs from other databases. Related herbs for triptolide were integrated from multiple TCM databases. The related targets and diseases for triptolide were derived from reference mining, database mining, or statistical inference, which are each displayed separately under different subpages. And help information about the data source is shown for each table. Among these, the records from reference mining, i.e. manual curation, give the highest-confidence data and can be cross-referenced into the reference page of HERB. The records from database mining are also of high quality, with the original database name and links provided. The significance of the records from statistical inference, i.e. Fisher's exact, is provided, in both P-values and FDRs followed by BH adjustment.

The high-throughput data analyzed by HERB is visualized in both tables and plots. Users can rank and filter records according to each column in all tables. We introduced four figures in the website to better visualize the data in these tables, with each representing an important aspect of the high-throughput data (Figure 3E). Note that human and mouse data are displayed separately, and users can navigate between the two datasets using drop-down menus. For example, the first volcano plot, shown in the upper left, visualizes the DEGs related to triptolide, which has been merged from multiple EXPs of triptolide. One dot in this figure represents a gene, with the color indicating the direction (up or down) of its dysregulation, and the size is proportional to the number of EXPs that support the gene as a DEG. The bar plot shown in the lower left and the dot plot shown in the upper right show the enriched GO terms and KEGG pathways. Each term in the underlying tables can be linked back to the GO and KEGG pages (48–50). Moreover, drop-down menus were provided for users to navigate GO/KEGG results for all, up, and down-regulated genes. The three colors in the GO figure represents three types of GO terms, molecular function (MF), cellular component (CC), and biological process (BP). The size of dots in the KEGG figures indicates the number of EXPs that support that particular item. Detailed information for each bar or dot are visible when the mouse is moved over the element. The last figure in the lower right is a network showing the modern compounds (in CMap) that are mappable to triptolide. The node in the center is triptolide. Other nodes

around it are related compounds in four different colors that stand for the data type of the CMap perturbation, including compound (CP), perturbational Class (PCL), gene over-expression (OE), gene knock-down (KD). Furthermore, we embedded a pie chart in each surrounding node to show the specific EXPs that support its mapping to triptolide. Edges between triptolide and its mapped compounds are shown in two colors, with red for a positive correlation and blue for a negative correlation between gene expression profiles. The widths of the edges are proportional to the absolute value of connectivity score. By default, we visualized mapped results with an absolute connectivity score >95. Users are free to select different cutoffs and subsets of the data types of CMap to focus on specific data of interest. And users can use the zoom in/out buttons at the upper-left to control the connectivity figure.

DISCUSSION

Transcriptome analysis allows quantitative measurements of the transcriptional responses and regulatory changes resulting from the perturbations due to compounds, making it a powerful approach for discovering drug targets, evaluating the therapeutic efficacy of drugs, and revealing possible side effects. In recent years, high-throughput data from laboratory or clinical studies of TCM herbs and ingredients have quickly accumulated, but to date there has been no good system of organization. Besides, new TCM-related references published within the last decade have not been curated. As a result, in this work, we reanalyzed all available microarray and RNA-seq data for TCM herbs/ingredients, and curated high-confidence targets and diseases information from recently published TCM references. Together with database mining and statistical inference, we ultimately built HERB, the most comprehensive database for TCM available to date.

The novelty of the HERB database includes: (i) HERB provides a comprehensive and unified pharmacotranscriptomics database of TCM, by reanalyzing all available high-throughput experiments for TCM. Using HERB, researchers and drug developers can view primary data as well as the data-driven mapping results between TCM herbs/ingredients and modern compounds, allowing easy exploration of potential mechanism of actions for herbs/ingredients as well as the identification of new potentially effective therapeutics. (ii) HERB gives high-confidence targets and diseases information related TCM herbs/ingredients based on manual curation of novel references published within the last decade, which bridging the large gap since the creation of HIT (2011) and TCMID (2012). Using these newly curated references in HERB, users can easily search and rank promising herbs or ingredients for a disease with a set of candidate targets. We believe this intuitive workflow will help researchers make use of the significant volume of published data related to TCM. (iii) HERB provides the most comprehensive list of herbs, ingredients, targets, and diseases by integrating multiple data resources. Further, HERB gives comprehensive pairwise relationships among them by combing diverse strategies including database mining, reference mining, and statistical inference.

HERB 提供了一个便利的网页界面，供用户浏览、搜索、可视化和下载关键信息，我们相信 HERB 将会大力支持中医现代化，指导药物研发。然而，目前可用的中医药药理转录组学数据集仍然不足，不仅在数据集数量上，而且在测序技术的多样性上。未来，我们计划持续添加尽可能多的数据。特别是，我们希望向 HERB 添加新的数据类型，如蛋白质组学、代谢组学和宏基因组学数据集，以进一步改进。随着更多多样化的数据变得可用，基于这些数据的进一步研究将是分析药材成分扰动下的基因调控网络 (51,52)，以及识别新的疾病相关基因类型 (53,54) 所必需的。总之，我们计划不断完善 HERB 数据库，为该领域提供高质量的资源。

补充数据

补充数据可在 NAR 在线获取。

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HERB provides a convenient web interface for users to browse, search, visualize and download key information, and we believed that HERB will intensively support the modernization of TCM to guide drug discovery efforts. However, the currently available data set for TCM pharmacotranscriptomics is remain insufficient, not only in the number of data sets, but also in the diversity of sequencing technologies. In the future, we plan to continuously add as much data as possible. In particular, we would like to add new types of data, such as proteomic, metabolomic, and meta-genomic datasets, to HERB for further improvements. With more and diverse data becoming available, further studies based on these data will be required for analyzing the gene regulation networks under the herb/ingredient perturbation (51,52), and identifying new types of disease-relevant genes (53,54). In a word, we plan to continuously improve the HERB database to provide a high-quality resource for the field of TCM big data.

SUPPLEMENTARY DATA

Supplementary Data are available at NAR Online.

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